

1. INTRODUCTION AND MAIN RESULTS

CONCLUSIONS

2. PRELIMINARIES

Definition 2.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

Definition 2.2 (Central factorials). *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n \left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1} \left(n + \frac{k}{2} - j\right)$ are falling factorials.

Proposition 2.3. (Central Newton's formula [?, p. 462]). *For arbitrary integers x , and t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes central finite difference

$$\delta^k f(t) = \sum_{j=0}^k (-1)^j \binom{k}{j} f\left(t + \frac{k}{2} - j\right).$$

Lemma 2.4 (Binomial form of central factorials). *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k(k-1)!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

Lemma 2.5 (Halved central factorials). *For integers $n, k \geq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \binom{n + \frac{k}{2}}{k} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k}.$$

Proof. We have $\binom{a}{k} = \frac{a}{k} \binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\binom{n + \frac{k}{2}}{k} = \frac{n + \frac{k}{2}}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Thus,

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \binom{n + \frac{k}{2}}{k} - \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \left[2 \binom{n + \frac{k}{2}}{k} - \binom{n + \frac{k}{2} - 1}{k-1} \right] = \frac{1}{2} \left[\binom{n + \frac{k}{2}}{k} + \binom{n + \frac{k}{2} - 1}{k} + \binom{n + \frac{k}{2} - 1}{k-1} - \binom{n + \frac{k}{2} - 1}{k-1} \right].$$

Thus, the statement follows. $\square$

Proposition 2.6 (Central Newton's formula for powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$n^m = \sum_{k=0}^m \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Proposition 2.7 (Halved central factorial powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n-t + \frac{k}{2}}{k} + \binom{n-t + \frac{k}{2} - 1}{k} \right] \delta^k t^m$$

Proposition 2.8 (Generalized hockey-stick identity). *For integers a, b and j ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right)$. By hockey stick identity $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^b \binom{k}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. $\square$

Lemma 2.9 (Centered hockey-stick identity). *For integers $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $j = 1 - t + \frac{k}{2} + r$, and $b = n - t + \frac{k}{2} + r$ yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k}.$$

By Generalized hockey-stick identity (2.8),

$$\sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Thus, the claim follows. □

Proposition 2.10 (Ordinary sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \delta^k t^m \sum_{j=1}^n \left[\binom{j-t+\frac{k}{2}}{k} + \binom{j-t+\frac{k}{2}-1}{k} \right]$$

Proposition 2.11 (Ordinary sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] \delta^k t^m$$

Proposition 2.12 (Central factorial numbers sums). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n+\frac{k}{2}+1}{k+1} - \binom{1+\frac{k}{2}}{k+1} + \binom{n+\frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1} \right] k! T(m, k),$$

where $T(n, k) = \frac{\delta^k 0^n}{k!}$ denote Central factorial numbers of the second kind (in Riordan's notation, see [?, p. 213], and [?].), defined by central Newton's formula for powers, evaluated

in $t = 0$

$$n^m = \sum_{k=0}^{\infty} T(m, k) n^{[k]}.$$

Definition 2.13 (Generalized Central factorial numbers of the second kind). *For integers $0 \leq k \leq m$, and an arbitrary integer t ,*

$$T_t(m, k) = \frac{\delta^k t^m}{k!},$$

defined by central Newton's formula for powers, evaluated at arbitrary integer t

$$n^m = \sum_{k=0}^{\infty} T_t(m, k) (n - t)^{[k]}.$$

Proposition 2.14 (Generalized Central factorial sums). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n - t + \frac{k}{2} + 1}{k+1} - \binom{1 - t + \frac{k}{2}}{k+1} + \binom{n - t + \frac{k}{2}}{k+1} - \binom{-t + \frac{k}{2}}{k+1} \right] k! T_t(m, k)$$

Proposition 2.15 (Double sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^2 n^m = \frac{1}{2} \sum_{k=0}^m \left[+ \binom{n - t + \frac{k}{2} + 2}{k+2} - \binom{2 - t + \frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{1 - t + \frac{k}{2}}{k+1} \Sigma^1 n^0 \right. \\ \left. + \binom{n - t + \frac{k}{2} + 1}{k+2} - \binom{1 - t + \frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{0 - t + \frac{k}{2}}{k+1} \Sigma^1 n^0 \right] \delta^k t^m \end{aligned}$$

Proposition 2.16 (Triple sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^3 n^m = \frac{1}{2} \sum_{k=0}^m \left[+ \binom{n - t + \frac{k}{2} + 3}{k+3} - \binom{3 - t + \frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{2 - t + \frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{1 - t + \frac{k}{2}}{k+1} \Sigma^2 n^0 \right. \\ \left. + \binom{n - t + \frac{k}{2} + 2}{k+3} - \binom{2 - t + \frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{1 - t + \frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{0 - t + \frac{k}{2}}{k+1} \Sigma^2 n^0 \right] \delta^k t^m \end{aligned}$$

Proposition 2.17 (Multifold sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k t^m \end{aligned}$$

Lemma 2.18 (Multifold sum of zero powers). *For integers $r \geq 0$ and $n \geq 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}$$

Proof. (1) Let $r = 0$, then $\Sigma^0 n^0 = n^0 = \binom{n-1}{0} = 1$, by definition (??).

(2) Let $r = 1$, then $\Sigma^1 n^0 = \sum_{k=1}^n \binom{k-1}{0} = \sum_{k=1}^n 1 = \binom{n}{1}$.

(3) Let $r = 2$, then $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \sum_{k=1}^n k = \binom{n+1}{2}$.

(4) Let $r = 3$, then $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$.

(5) By induction over r and hockey stick identity $\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}$, the claim follows

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

□

Proposition 2.19 (Multifold binomial sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k t^m \end{aligned}$$

Proposition 2.20 (Multifold central factorial sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} k! T_t(m, k) \end{aligned}$$

Definition 2.21.

$$R(n, t, k, r) = \binom{n - t + \frac{k}{2} + r}{k + r} + \binom{n - t + \frac{k}{2} + r - 1}{k + r} \\ - \sum_{s=0}^{r-1} \left[\binom{r - s - t + \frac{k}{2}}{k + r - s} + \binom{r - 1 - s - t + \frac{k}{2}}{k + r - s} \right] \Sigma^s n^0$$

Proposition 2.22 (Multifold sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^r n^m = \sum_{k=0}^m \frac{\delta^k t^m}{2} R(n, t, k, r)$$

Example 2.23 (Sums of squares).

$$\begin{aligned} t = 0 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(1 + n) + \frac{\delta^2 t^2}{2} \frac{n(1 + 3n + 2n^2)}{6}, \\ t = 1 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n - 1) + \frac{\delta^2 t^2}{2} \frac{n(1 - 3n + 2n^2)}{6}, \\ t = 2 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n - 3) + \frac{\delta^2 t^2}{2} \frac{n(13 - 9n + 2n^2)}{6}, \\ t = 3 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n - 5) + \frac{\delta^2 t^2}{2} \frac{n(37 - 15n + 2n^2)}{6}, \\ t = 4 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n - 7) + \frac{\delta^2 t^2}{2} \frac{n(73 - 21n + 2n^2)}{6}. \end{aligned}$$

Example 2.24 (Sums of cubes).

$$\begin{aligned} t = 0 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(1 + n) + \frac{\delta^2 t^3}{2} \frac{n(1 + 3n + 2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-1 + n + 4n^2 + 2n^3)}{24}, \\ t = 1 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n - 1) + \frac{\delta^2 t^3}{2} \frac{n(1 - 3n + 2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(1 + n - 4n^2 + 2n^3)}{24}, \\ t = 2 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n - 3) + \frac{\delta^2 t^3}{2} \frac{n(13 - 9n + 2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-21 + 25n - 12n^2 + 2n^3)}{24}, \\ t = 3 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n - 5) + \frac{\delta^2 t^3}{2} \frac{n(37 - 15n + 2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-115 + 73n - 20n^2 + 2n^3)}{24}, \\ t = 4 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n - 7) + \frac{\delta^2 t^3}{2} \frac{n(73 - 21n + 2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-329 + 145n - 28n^2 + 2n^3)}{24}. \end{aligned}$$

Example 2.25 (Double sum of squares).

$$t = 0 : \quad \Sigma^2 n^2 = \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(1+n)^2(2+n)}{12},$$

$$t = 1 : \quad \Sigma^2 n^2 = \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-1+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n^2(-1+n^2)}{12},$$

$$t = 2 : \quad \Sigma^2 n^2 = \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-4-3n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(10+5n-4n^2+n^3)}{12},$$

$$t = 3 : \quad \Sigma^2 n^2 = \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-7-6n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(32+23n-8n^2+n^3)}{12},$$

$$t = 4 : \quad \Sigma^2 n^2 = \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-10-9n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(66+53n-12n^2+n^3)}{12}.$$

Example 2.26 (Triple sum of squares).

$$t = 0 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(6+11n+6n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(18+45n+40n^2+15n^3+2n^4)}{120},$$

$$t = 1 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-2-n+2n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(-2-5n+5n^3+2n^4)}{120},$$

$$t = 2 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-10-13n-2n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(58+65n-5n^3+2n^4)}{120},$$

$$t = 3 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-18-25n-6n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(198+255n+40n^2-15n^3+2n^4)}{120},$$

$$t = 4 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-26-37n-10n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(418+565n+120n^2-25n^3+2n^4)}{120}.$$

Lemma 2.27 (Stirling form of Binomial). *For integers n, k*

$$\binom{n}{k} = \sum_{j=0}^k \frac{(-1)^{k-j}}{k!} \begin{bmatrix} k \\ j \end{bmatrix} n^j$$

Example 2.28 (Examples of multifold binomial sums).

$$\begin{aligned}
t = 0 : \quad \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r} \right. \\
&\quad \left. - \sum_{s=0}^{r-1} \left[\binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 0^m, \\
t = 1 : \quad \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r} \right. \\
&\quad \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-1+\frac{k}{2}}{k+r-s} + \binom{r-2-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 1^m, \\
t = 2 : \quad \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-2+\frac{k}{2}+r}{k+r} + \binom{n-2+\frac{k}{2}+r-1}{k+r} \right. \\
&\quad \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-2+\frac{k}{2}}{k+r-s} + \binom{r-3-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 2^m.
\end{aligned}$$

t	$T_t(2m, 2k)$	OEIS
0	1, 0, 1, 0, 1, 1, 0, 1, 5, 1, 0, 1, 21, 14, 1, 0, 1, 85, 147, 30, 1, $\dots$	A269945
1	1, 1, 1, 1, 7, 1, 1, 31, 20, 1, 1, 127, 231, 42, 1, 1, 511, $\dots$	A000000
2	1, 4, 1, 16, 25, 1, 64, 301, 65, 1, 256, 3025, 1701, 126, 1, $\dots$	A000000
3	1, 9, 1, 81, 55, 1, 729, 1351, 140, 1, 6561, 26335, 6951, 266, 1, $\dots$	A000000

3. MATHEMATICA

Mathematica Function	Validates / Prints
<code>MultifoldSumOfPowersRecurrence[r, n, m]</code>	Computes $\Sigma^r n^m$, (2.1)
<code>CentralFactorial[n, k]</code>	Computes $n^{[k]}$, (2.2)
<code>ValidateHalvedCentralBinomials[20]</code>	Validates Lem. (2.5)
<code>ValidateNewtonsFormulaForPowers[10]</code>	Validates Prop. (2.6)
<code>ValidateHalvedCentralFactorialPowers[5]</code>	Validates Prop. (2.7)
<code>ValidateCenteredHockeyStickIdentity[5]</code>	Validates Lem. (2.9)
<code>ValidateOrdinarySumsOfPowersDecomposition[10]</code>	Validates Prop. (2.11)
<code>ValidateDoubleSumsOfPowersDecomposition[5]</code>	Validates Prop. (2.15)
<code>ValidateTripleSumsOfPowersDecomposition[5]</code>	Validates Prop. (2.16)
<code>ValidateMultifoldSumsOfPowers[3]</code>	Validates Prop. (2.17)